US012415093B2

(12) United States Patent
Perkins et al.

(10) Patent No.: US 12,415,093 B2
(45) Date of Patent: Sep. 16, 2025

(54) RADIOTHERAPY APPARATUS

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 293 days.

(21) Appl. No.: **18/061,258**

(22) Filed: **Dec. 2, 2022**

(65) **Prior Publication Data**

US 2024/0181276 A1 Jun. 6, 2024

(30) **Foreign Application Priority Data**

Dec. 6, 2021 (GB) 2117574

(51) **Int. Cl.**
A61N 5/10 (2006.01)

(52) **U.S. Cl.**
CPC ***A61N 5/1081*** (2013.01); ***A61N 5/1048*** (2013.01)

(58) **Field of Classification Search**
CPC A61N 5/1048; A61N 5/1081; A61N 5/01; A61N 5/10; A61N 5/1001; A61N 5/1064; A61N 5/1065; A61N 2005/1022; A61N 2005/1092; A61N 5/1077; A61N 5/1082; A61N 2005/1057; A61N 2005/005; A61N 5/1084; A61N 2005/1095; A61N 5/1049; A61N 2005/1061; A61N 5/1042; A61N 5/1075; A47B 91/02; A61B 6/4447; A61B 6/4488; A61B 6/032; A61B 6/4435; G01N 3/04
USPC 378/4, 15, 196
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,823,037 B2 * 11/2004 Riemer G01N 23/046 378/197
6,840,673 B2 * 1/2005 Moritake A61B 6/035 378/198
6,959,068 B1 * 10/2005 Sommer A61B 6/04 378/208

(Continued)

FOREIGN PATENT DOCUMENTS

EP	1738798	1/2007
EP	2865417	4/2015
GB	2519592 A	4/2015

OTHER PUBLICATIONS

"European Application Serial No. 22211204.7, European Search Report dated May 9, 2023", (May 9, 2023), 7 pgs.

(Continued)

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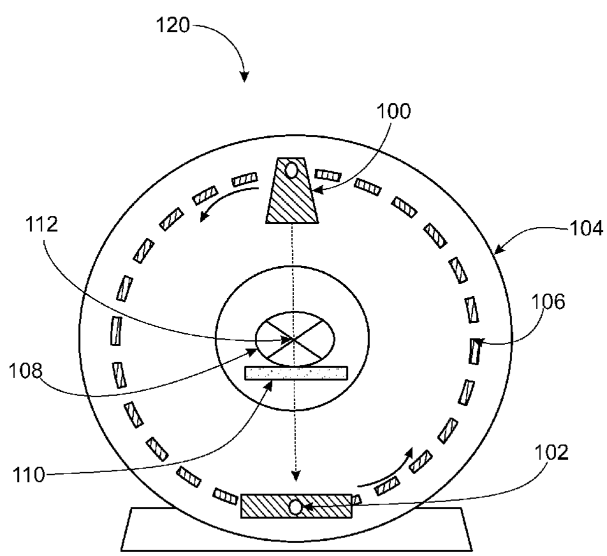
(57) **ABSTRACT**

Disclosed herein is a radiotherapy apparatus comprising a gantry rotatable about a gantry rotation axis, and a plurality of rotatable rollers positioned underneath the gantry and configured to support the gantry. A first rotatable roller of the plurality of rotatable rollers is configured to rotate about a first axle, and the first axle is configured to rotate about a first axle rotation axis. The first axle is configured such that rotation of the first axle about the first axle rotation axis displaces the first rotatable roller in a direction substantially perpendicular to the gantry rotation axis.

18 Claims, 9 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

2004/0062343	A1*	4/2004	Brunnett	A61B 6/035 378/4
2009/0139344	A1	6/2009	Lindeman	

OTHER PUBLICATIONS

"British Application No. 2117574.0, Office Action dated May 27, 2022", (May 27, 2022), 5 pgs.

* cited by examiner

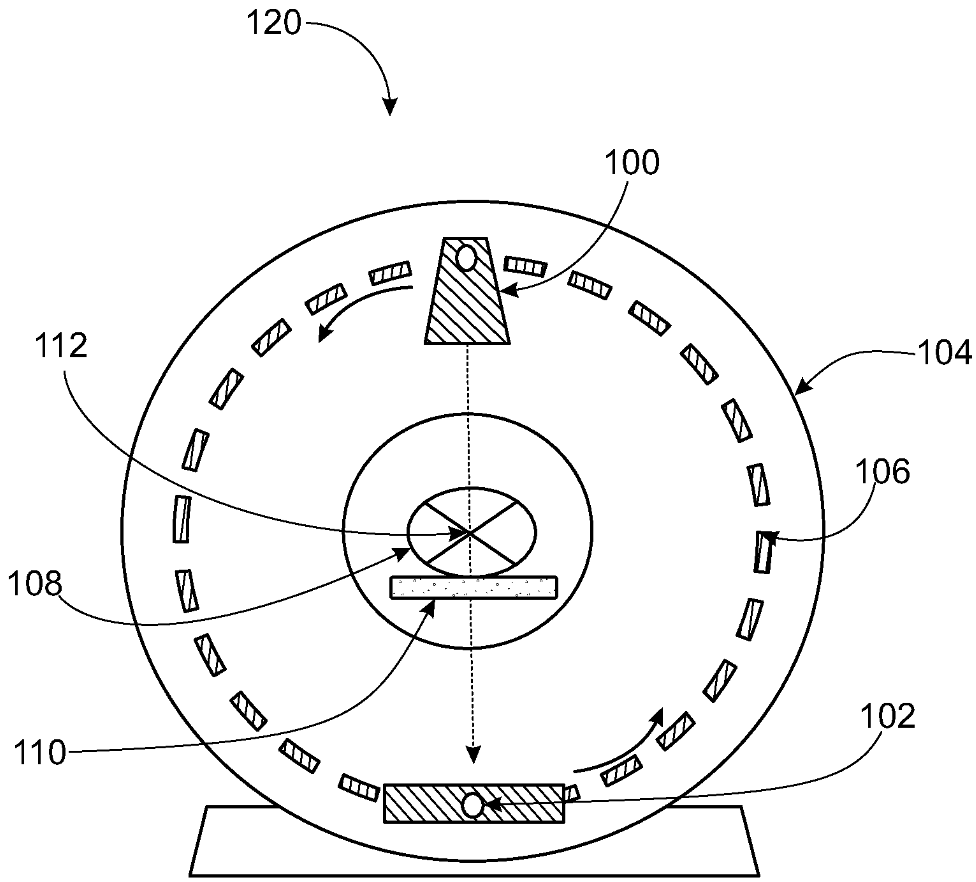


FIG. 1

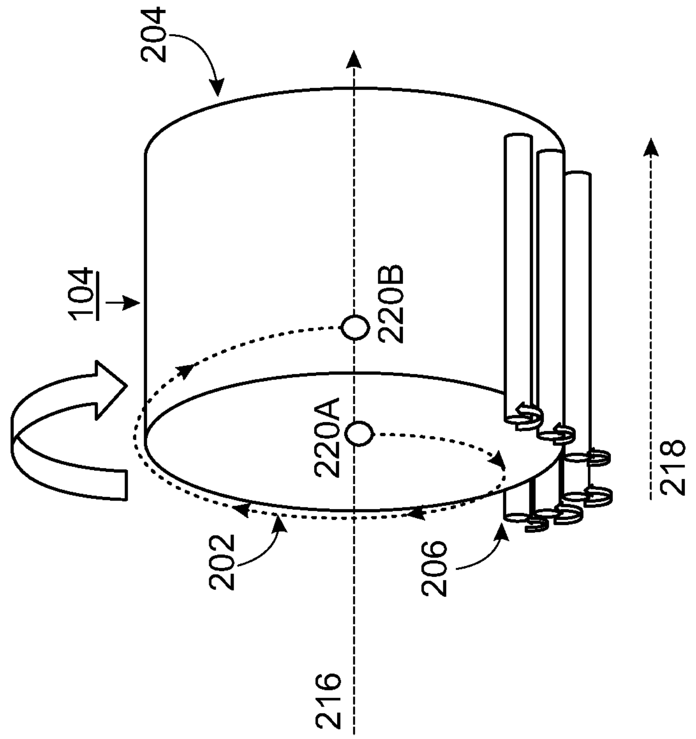


FIG. 2B

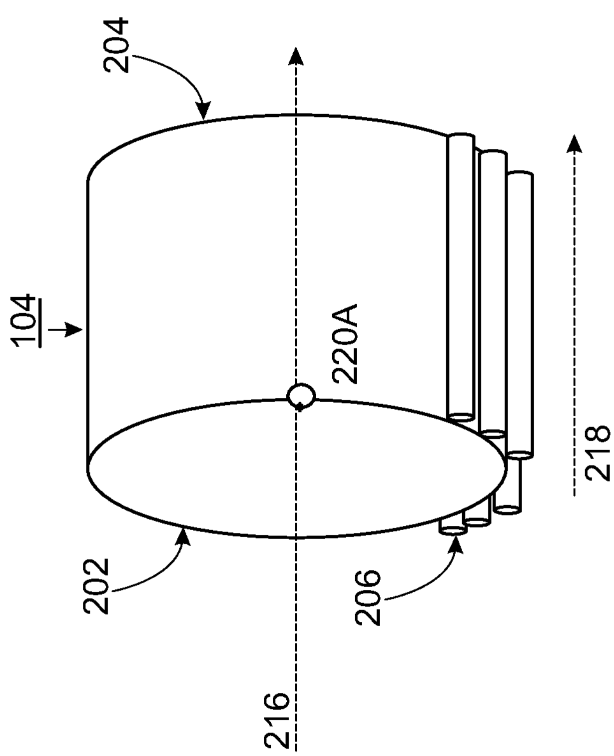


FIG. 2A

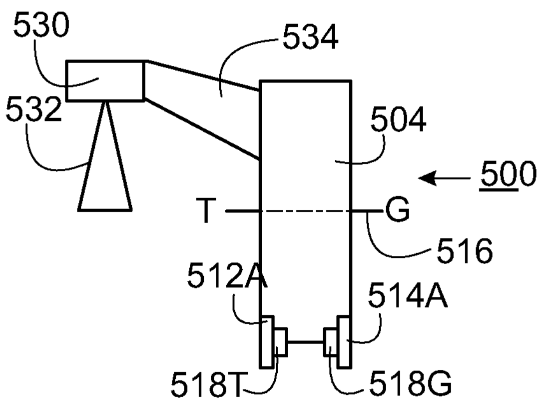


FIG. 3A

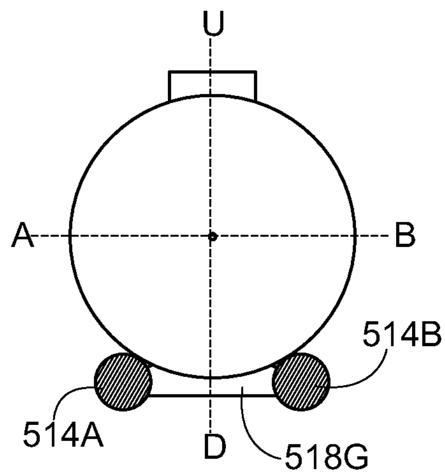


FIG. 3B

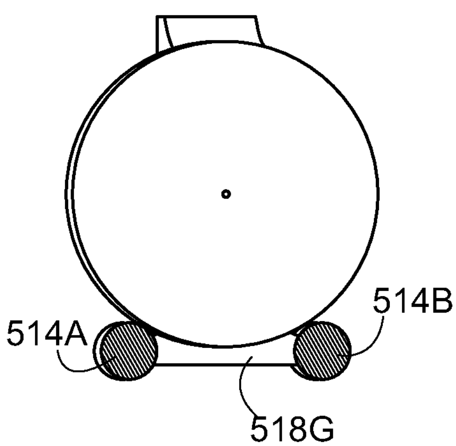


FIG. 3C

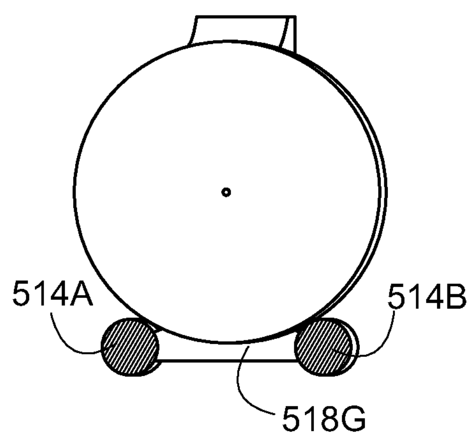


FIG. 3D

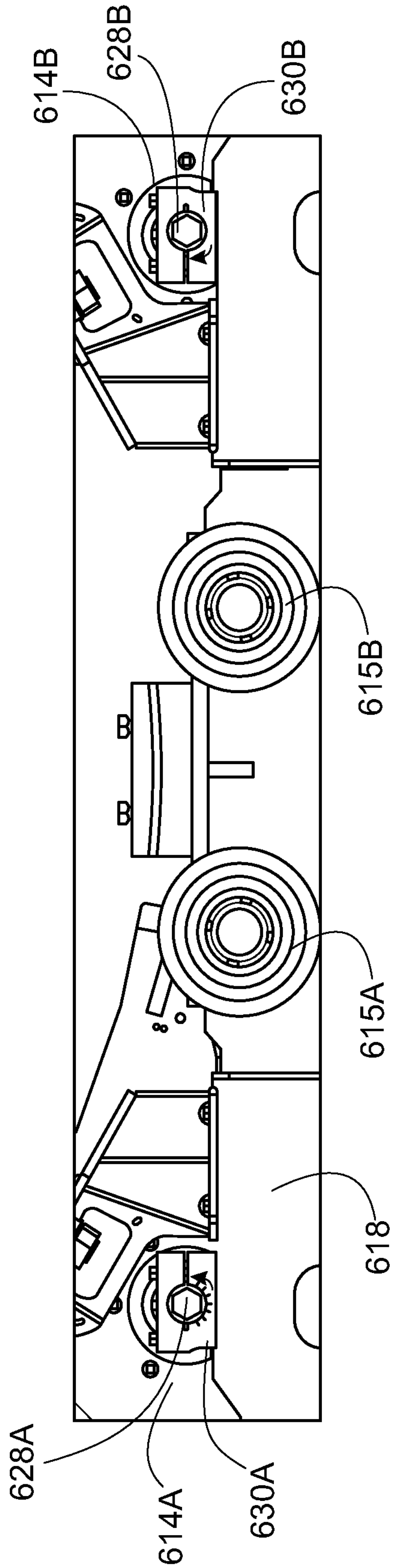


FIG. 4A

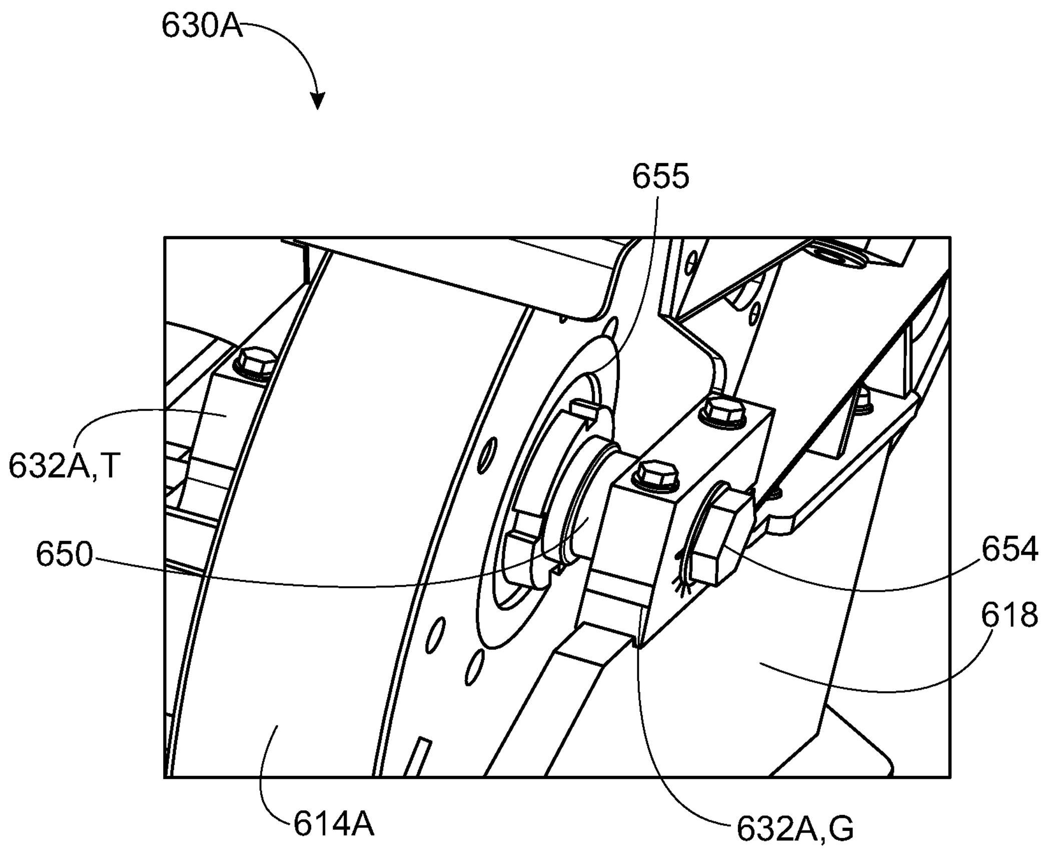


FIG. 4B

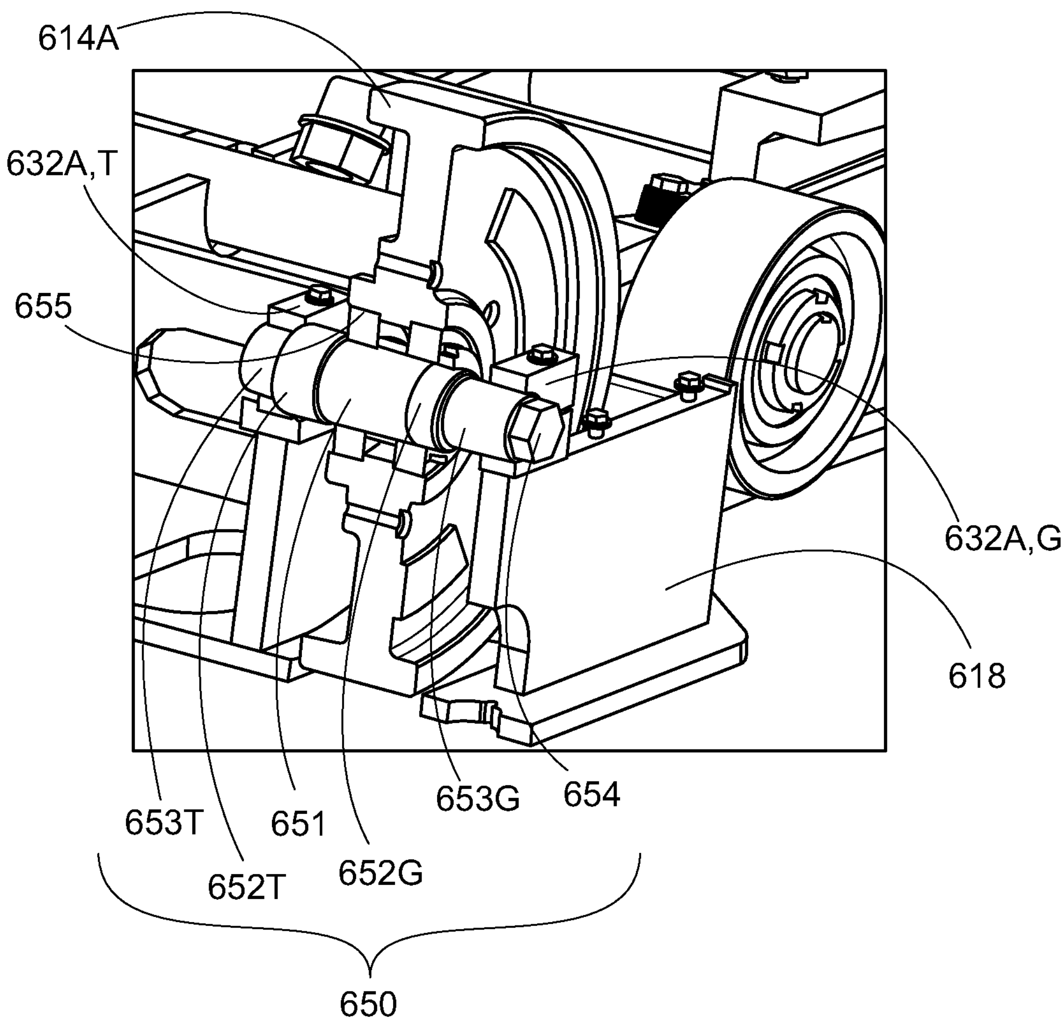


FIG. 5

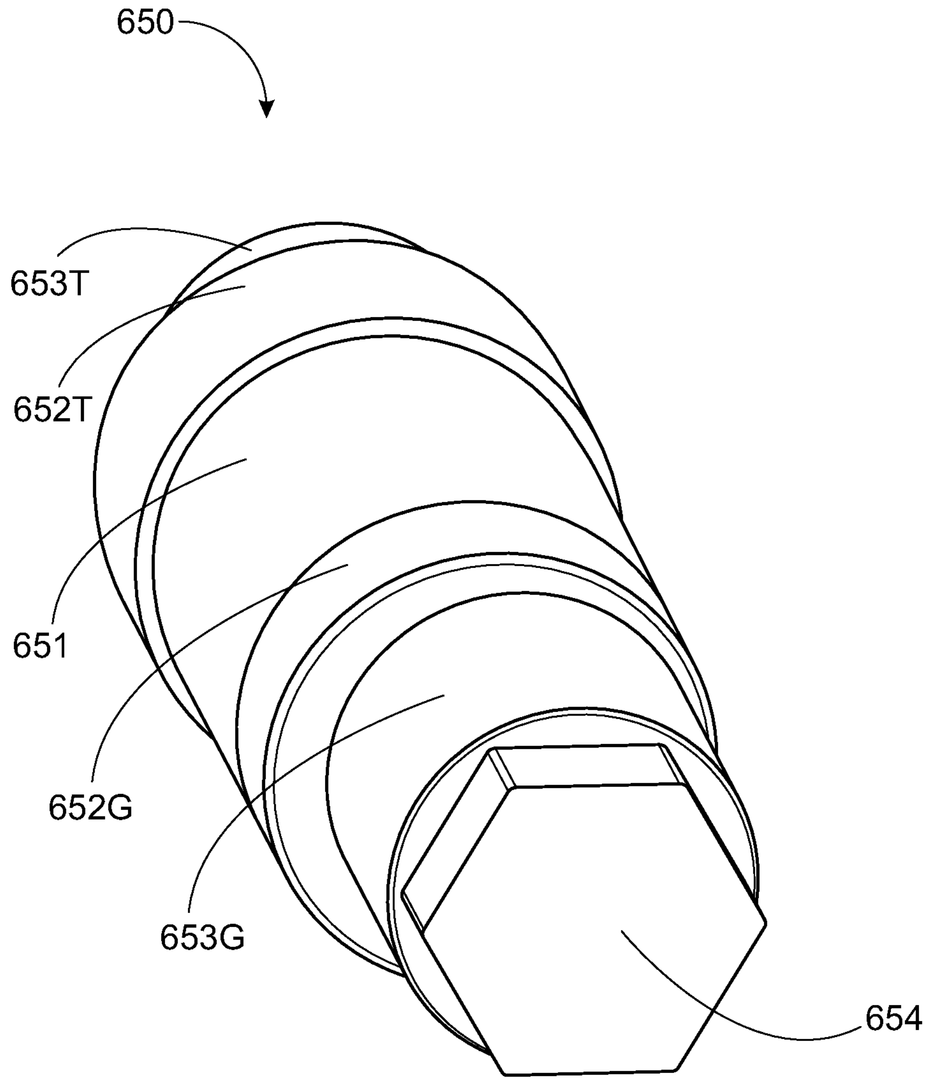


FIG. 6

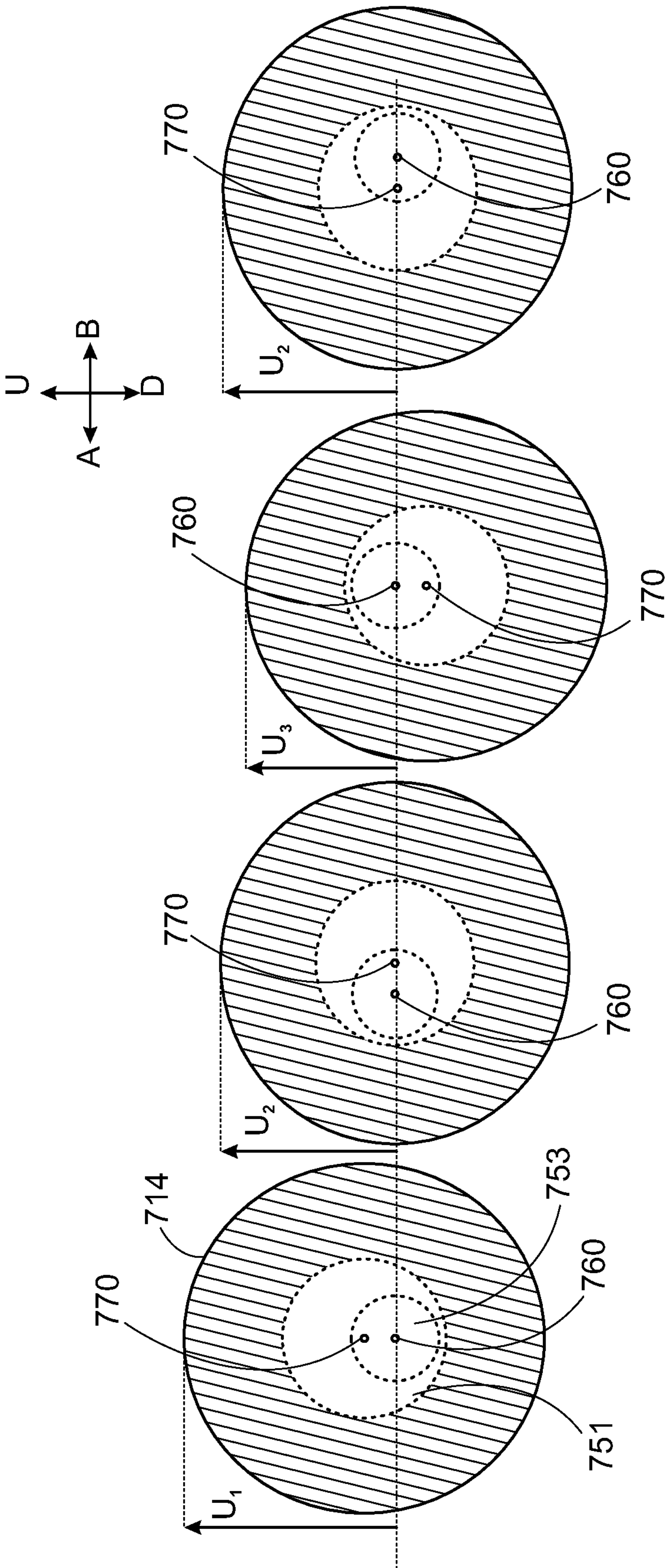


FIG. 7D

FIG. 7C

FIG. 7B

FIG. 7A

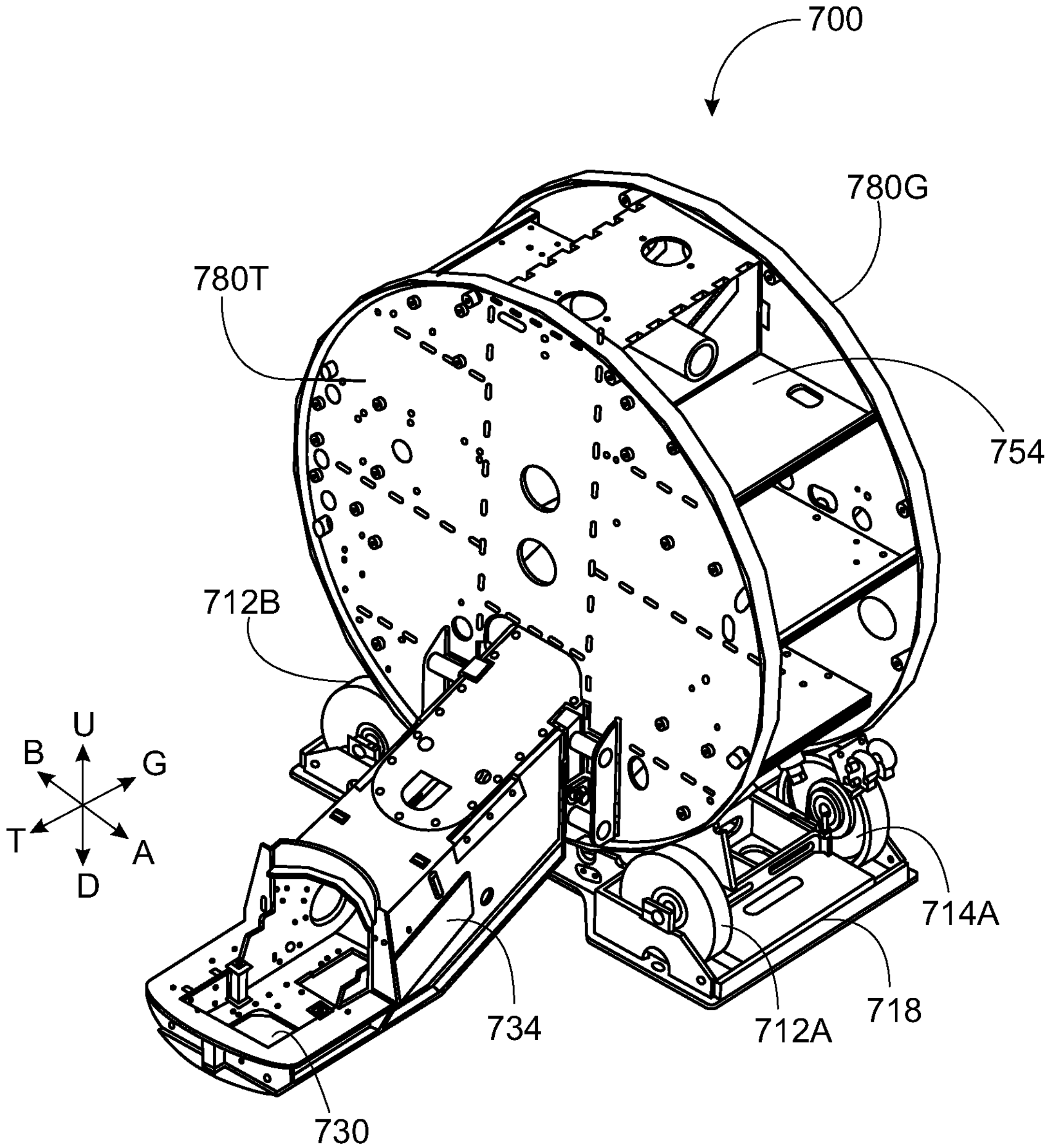


FIG. 8

RADIOTHERAPY APPARATUS

CLAIM FOR PRIORITY

This application claims the benefit of priority of United Kingdom Application No. 2117574.0, filed Dec. 6, 2022, which is hereby incorporated by reference in its entirety.

This disclosure relates to a radiotherapy apparatus and method, and in particular to an apparatus comprising components and/or means for mitigating or eliminating the 'corkscrew' effect.

BACKGROUND

Radiotherapy can be described as the use of ionizing radiation, such as X-rays, to treat a human or animal body. Radiotherapy is commonly used to treat tumors within the body of a patient or subject. In such treatments, ionizing radiation is used to irradiate, and thus destroy or damage, cells which form part of the tumor.

For the purposes of radiotherapy treatment, it is desirable to deliver a particular dose to a target region while minimizing the dose to surrounding areas of healthy tissue. Thus, the accuracy of the delivery of the treatment beam is of the utmost importance during radiotherapy treatment. Any inaccuracy in delivery of the treatment beam can negatively impact treatment efficacy, and may result in a higher dose being delivered to healthy tissue.

A radiotherapy device can comprise a gantry which supports a beam generation system, or other source of radiation, which is rotatable around a patient. For example, for a linear accelerator (linac) device, the beam generation system may comprise a source of radio frequency energy, a source of electrons, an accelerating waveguide, beam shaping apparatus, etc.

The gantry of a radiotherapy device may be provided as a rotatable drum with a circular cross-section. The gantry is rotatable about a gantry rotation axis. The beam generation system is coupled to the rotatable gantry, and therefore rotation of the gantry allows for the beam of therapeutic radiation to be applied from multiple angles around the patient during a treatment session. Radiation is deliverable toward a radiation isocenter, which may be positioned on the gantry rotation axis regardless of the angle to which the radiation source is rotated around the gantry. The gantry drum may be supported by a number of rollers, such as wheels. Devices such as rollers, and the like, can be present to support or drive the rotation of the gantry drum.

SUMMARY

It has been found that, due to forces generated during its rotation, a rotating gantry drum can be caused to translate along its axis of rotation. The gantry drum may become displaced in one direction along the rotational axis when the drum undergoes a rotation in a first direction, and the drum may return along substantially the same axial path when rotated in the opposite direction. This movement can be of the order of 1 mm or more. This translation of the gantry drum can be referred to as 'corkscrewing', and can be referred to as the 'corkscrew' effect.

Any movement of the gantry along its axis causes a shift in isocenter location, and thus affects the accuracy of treatment beam delivery. The forces generated by this 'corkscrewing' of the gantry drum can also bend or damage brackets holding rollers used to rotate the gantry in a radiotherapy apparatus.

Roller bearings can be used to attempt to mitigate the corkscrew effect, however, these roller bearings have a limited effect due to the large forces generated by the 'corkscrewing' and the significant mass of the gantry. These large forces generated by the corkscrewing can lead to damage of the components of the radiotherapy device.

According to an aspect of the present disclosure, there is provided a radiotherapy apparatus comprising a gantry rotatable about a gantry rotation axis, and a plurality of rotatable rollers positioned underneath the gantry and configured to support the gantry. A first rotatable roller of the plurality of rotatable rollers is configured to rotate about a first axle, and the first axle is configured to rotate about a first axle rotation axis. The first axle is configured such that rotation of the first axle about the first axle rotation axis displaces the first rotatable roller in a direction substantially perpendicular to the gantry rotation axis.

Optionally, the apparatus further comprises a support structure comprising a first axle holder configured to support the first axle and thereby define the first axle rotation axis.

Optionally, the first axle comprises different sections along its length, including an eccentric section, and the first roller is configured to rotate about the eccentric section.

Optionally, the eccentric section is positioned eccentrically with respect to the first axle rotation axis.

Optionally, the eccentric section is offset with respect to the first axle rotation axis.

Optionally, the eccentric section has a central axis which is parallel with, but which does not align with, the first axle rotation axis.

Optionally, the eccentric section defines a first roller rotation axis about which the first rotatable roller rotates, wherein the first roller rotation axis is displaced from the first axle rotation axis.

Optionally, the first axle is configured such that, by rotating the first axle about the first axle rotation axis in a first rotational direction, the first roller rotation axis is rotated in the first rotational direction.

Optionally, the first axle further comprises holding sections positioned either side of the eccentric section, wherein the holding sections pass through the axle holder.

Optionally, the holding sections are substantially cylindrical, and have a shared central axis which aligns with the first axle rotation axis.

Optionally, the plurality of rotatable rollers comprises at least one drive roller configured to drive rotation of the gantry.

Optionally, the plurality of rotatable rollers further comprises a second rotatable roller, wherein the second rotatable roller is configured to rotate about a second roller axle, wherein the second axle is configured to rotate about a second axle rotation axis, and wherein the second axle is configured such that rotation of the second axle about the second axle rotation axis displaces the second rotatable roller in a direction substantially perpendicular to the gantry rotation axis.

Optionally, the first and second axles are horizontally displaced with respect to each other along an axis substantially perpendicular to the gantry rotation axis.

Optionally, the first and second rotatable rollers are positioned either toward the rear, or toward the front, of the gantry.

According to another aspect of the present disclosure, there is provided a method for displacing a first roller of a plurality of rotatable rollers positioned underneath a gantry of a radiotherapy device. The rotatable rollers are configured to support the gantry, and the gantry is rotatable about a gantry rotation axis. The first rotatable roller of the plurality of rotatable rollers is configured to rotate about a first axle, and the first axle is configured to rotate about a first axle rotation axis. The first axle is configured such that rotation of the first axle about the first axle rotation axis displaces the first rotatable roller in a direction substantially perpendicular to the gantry rotation axis. The method comprises rotating the first axle about its rotation axis to displace the first rotatable roller in the direction substantially perpendicular to the gantry rotation axis.

The radiotherapy device for use with the method may be in accordance with one or more of the radiotherapy devices disclosed herein.

BRIEF DESCRIPTION OF THE DRAWINGS

In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. Like numerals having different letter suffixes may represent different instances of similar components. The drawings illustrate generally, by way of example, but not by way of limitation, various embodiments discussed in the present document.

FIG. **1** depicts a radiotherapy apparatus according to the present disclosure.

FIG. **2***a* and FIG. **2***b* depict a gantry drum arrangement experiencing a corkscrew effect.

FIGS. **3***a-d* depict a radiotherapy apparatus.

FIGS. **4***a* and **4***b* depict a support structure and a plurality of support rollers for supporting a gantry of a radiotherapy apparatus, such as the radiotherapy apparatus depicted in FIGS. **3***a-d*.

FIG. **5** depicts a cross-section through a rotatable roller and roller bearing of the apparatus depicted in FIGS. **4***a* and **4***b*

FIG. **6** depicts an axle comprising an eccentric section, suitable for use in the apparatus depicted in FIG. **4***a*, FIG. **4**B and FIG. **5**;

FIGS. **7***a-d* are schematic depictions of an axle suitable for use in the apparatus depicted in FIG. **4***a*, FIG. **4**B and FIG. **5**;

FIG. **8** depicts a radiotherapy apparatus in accordance with the present disclosure.

DETAILED DESCRIPTION

In overview, and without limitation, the present disclosure relates to radiotherapy apparatus comprising components and/or means for mitigating, or else eliminating, an undesirable corkscrewing effect. The gantry is supported by a plurality of rollers (e.g. wheels), which are in turn supported by an under-gantry support structure. At least one of the rollers can be displaced in a direction substantially perpendicular to the gantry rotation axis. Hence, the gantry's position on the supporting rollers, and the gantry's alignment with the rotational axes of the rollers, can be adjusted. By making this kind of adjustment, it is possible to mitigate the corkscrewing effect, or else find a 'sweet spot' in which the corkscrewing effect may be eliminated entirely. Accordingly, the radiotherapy device stability is improved, and in particular the device's axial stability during rotation of the device can be improved.

An individual roller can be adjusted by virtue of an axle comprising an eccentric section, where the roller is positioned and configured to rotate around (and with respect to) the eccentric section. The axle and eccentric section are configured such that rotation of the axle about its own rotational axis causes a displacement of the roller in a direction substantially perpendicular to the gantry rotation axis. Also disclosed herein is a method for displacing one or more rollers in a manner which mitigates or eliminates the corkscrewing effect.

FIG. **1** depicts a radiotherapy apparatus according to the present disclosure. The arrangement described should be considered as providing one or more examples of a radiotherapy apparatus **120** and it will be understood that other arrangements are possible. FIG. **1** shows a cross-section through a radiotherapy apparatus **120** comprising a radiation source **100** and a detector **102** attached to a gantry **104**. The gantry **104** may take the form of a substantially cylindrical drum. The radiation source **100** and the detector **102** are fixed to the gantry drum **104** so as to rotate with the gantry drum **104**. The radiation source **100** and the detector **102** are arranged diametrically opposite one another.

FIG. **1** also depicts a subject **108** on a support surface **110**. The support surface **110** may be moved longitudinally relative to the gantry drum **104** (i.e. away from the plane of the gantry drum **104**), for example to aid positioning of the subject **108**. In some examples, the support surface **110** may be moved along other translational axes (e.g. in the plane of the gantry) and/or rotational axes. A controller may have access to position and/or movement information for the support surface **110**, and may be configured to position a patient according to the requirements of a particular radiotherapy treatment plan. As radiation is applied to the subject **108**, for example according to a treatment plan, the radiation source **100** and the detector **102** rotate together with the gantry **104** such that they are always arranged 180° from one another. The radiation source **100** directs radiation towards the subject **108** from various angles around the subject **108** in order to spread out the radiation dose received by healthy tissue to a larger region of healthy tissue while building up a prescribed dose of radiation at a target region. As shown in FIG. **1**, radiation may be emitted in a plane which is perpendicular to the axis of rotation of the radiation source **100**. Thus, radiation may be applied to a radiation isocentre **112** at the centre of the gantry drum **104** regardless of the angle to which the radiation source **100** is rotated around the gantry drum **104**. The radiation can be shaped or limited using a collimator, for example a multi-leaf collimator, arranged in the path of a radiation beam.

FIG. **2**A is a simplified schematic diagram of a gantry. The gantry is a drum-shaped gantry as described above in relation to FIG. **1**. Such a gantry may form part of the radiotherapy apparatus **120** described with respect to FIG. **1**, though a number of components described in relation to FIG. **1** are not shown here for simplicity. The gantry itself may comprise additional components, such as an arm for connecting to a source of radiation. However, FIGS. **2**A and **2**B primarily depict the drum **104** of the gantry. The drum **104** has a circular cross section, and is substantially cylindrical in shape. The drum **104** is oriented in the vertical plane, i.e. the central axis of the drum **104** is substantially parallel with a horizontal axis and substantially perpendicular to a vertical axis. The drum **104** comprises a first end **202**, and a second end **204**, each comprising a circular cross-section. The first end **202** may be proximal to the subject **108** during treatment, and the radiation source **100** and the detector **102** may be fixed or coupled to the first end **202** of the gantry drum **104**, for example via an arm. The radiation source **100** and detector **102** rotate with the gantry drum **104** during treatment. The first end may also be referred to as the 'front' or 'T' end of the gantry. The second end **204** may be distal to the subject **108** during treatment, and may be connected to, for example, a power supply, a cooling system, and/or other components of the radiotherapy apparatus. The second end may also be referred to as the 'rear' or 'G' end of the gantry.

A number of rollers **206** support the drum **104** as it rotates. The rollers **206** may be substantially cylindrical as shown, and may run longitudinally along the underside of the drum. The rollers **206** substantially align along a roller axis **218**, which is substantially parallel to the gantry rotational axis **216**. One or more of the rollers are drive rollers. During use, the drive rollers are driven in order to cause the gantry drum **104**. Each roller **206** is rotatable around a parallel roller rotational axis unique to each roller **206**.

FIG. **2**B depicts the gantry drum during or directly after a rotation of the drum **104** has occurred. Numerous forces act upon the drum **104** during its rotation. Such forces include rotational forces imparted by the drive rollers, forces due to the effect of gravity on the drum **104**, frictional forces between rollers **206** and the rotating gantry drum **104**, and the like. These forces do not act on the gantry **104** in a uniform manner, and may introduce an imbalance of forces. For example, gravitational forces act on the outer surfaces of the drum **104** to mechanically deform them, and because heavy components are attached to the gantry at different parts around its circumference, this deformation is variable depending on the angle of gantry rotation.

An imbalance of forces, mechanical tolerances, motion, and mechanical deformations caused by gravity can all contribute to introducing an angle between the gantry rotational axis **216** and the roller axis **218**, which may in turn cause the drum **104** to 'steer' in one direction or another. For example, the drum **104** may move longitudinally along its rotational axis, as depicted in FIG. **2**B. In FIG. **2**B, the drum **104** is displaced in a direction parallel to arrow **217**. In other implementations, the drum **104** may be displaced in the opposition direction to arrow **217**. The direction of this longitudinal displacement may depend on the direction of rotation of the gantry drum. For example, a gantry drum may rotate in a clockwise direction and become displaced in a first direction, and may rotate in an anti-clockwise direction and become displaced in a second direction, opposite to the first direction. This is known as the 'corkscrewing' of the gantry, or the 'corkscrew' effect, due to the similarity between the rotational displacement of the gantry and the movement caused by the twisting of a corkscrew.

The 'corkscrew'-like motion of the gantry can be observed by following a point **220**A on the surface of the gantry drum. The point is defined relative to the gantry rotational axis **216** in the longitudinal direction. As the drum rotates, the point **220**A is displaced, from its initial position, longitudinally along the gantry rotational axis **216**. As depicted by the trajectory **222** on FIG. **2**B, the point **220**A follows a helical path as the drum **104** rotates. At some later time, the point on the drum is no longer at location **220**A defined relative to the gantry rotational axis **216**, but is at location **220**B. This displacement is merely shown to illustrate the displacement of the gantry, and is not to scale.

Because the gantry is configured to support the radiation source **100** and detector **102**, any displacement of the gantry due to the corkscrewing effect will result in a displacement of the radiation source **100** and the detector **102**. Such a displacement of the radiation source **100** can cause small shifts in isocenter **112** location, and thus affect the efficacy and accuracy of treatment beam delivery. A displacement of the gantry drum **104** in the manner of FIG. **2**B may reduce the efficacy of treatment by reducing the accuracy with which the therapeutic treatment beam is applied.

FIGS. **3***a-d* depicts a radiotherapy apparatus **500**. The apparatus **500** comprises a gantry **504** in the form of a drum. The gantry **504** comprises an axis of rotation **516**.

The apparatus **500** comprises a source of radiation **530** which is coupled to the gantry via an arm **534**. FIG. **3***a* schematically depicts a beam of radiation **532** being emitted from the source of radiation **530**. The source of radiation **530** is part of a beam generation system which comprises an electron gun, a waveguide to accelerate electrons emitted by the electron gun toward a target, and a target which, when struck by high energy electrons, produces high energy X-rays.

The gantry rotation axis **516** lies along a T-G axis, where the G refers to the (electron) gun end of the waveguide and the T refers to the target end of the waveguide in an implementation in which the waveguide is comprised within the arm **534**. The use of this terminology for the T-G axis is merely used to describe the geometry of the radiotherapy apparatus **500**, and does not imply that the waveguide need be oriented in this way in implementations of the present invention. The T-G axis passes through the isocenter and is horizontal when the apparatus is viewed from the view depicted in FIG. **3***a*. The 'front' of the device **500**, i.e. the side on which the source of radiation **530** is positioned, can be described as the T end of the device **500**. Similarly, the rear of the device **500** can be described as the G end of the device **500**. The geometry of the device **500** can be further described with respect to an A-B axis, which is depicted in FIG. **3***b*. The A-B passes through the isocenter, is orthogonal with the T-G axis, and is horizontal when the apparatus is viewed from the view depicted in FIG. **3***b*. Finally, the apparatus can be described with respect to a U-D (up-down) axis. The U-D axis also passes through the isocenter and is orthogonal with the T-G and A-B axes. Where appropriate for improving clarity of description, certain components herein may be referred to using these axes, and certain reference numerals herein are appended with letters G, T, A, B, U, D, particularly where the component is part of a plurality of similar components and it is useful to distinguish between the components by reference to the apparatus geometry.

The gantry **504** is supported on a plurality of rotatable rollers **512**A,B and **514** A,B. The rollers are substantially cylindrical, and may take the form of wheels or elongated cylinders. Herein, the term "rollers" will be used to describe rotatable cylinders on which the gantry is supported, and the term rollers should be understood to describe wheel-shaped rollers (as shown in FIGS. **3***a-d*) and also cylinders with a longer central axis than might be typically associated with a "wheel" shape.

Each of the rollers **512**A,B and **514** A,B is positioned underneath the gantry **504** and is configured to support the gantry **504**. The rollers **512**A,B and **514** A,B are each in contact with the gantry **504**. The outer circumferences of the rollers **512**A,B and **514** A,B are in contact with an outer circumference of the gantry **504**. The rollers **512**A,B and **514** A,B are rotationally coupled with the gantry **504** such that the rollers **512**A,B and **514** A,B and the gantry **504** rotate together about their respective rotational axes.

One or more of the plurality of rollers **512**A,B and **514** A,B may be a drive roller. In some implementations, all of the rollers **512**A,B and **514** A,B are drive rollers. The one or more drive rollers are coupled with a motor or other drive means. The one or more drive rollers enable rotation of the gantry **504** by actuation of the motor or other drive means. To rotate the gantry, the motor(s) or other drive means control rotation of the drive rollers, which via friction causes the gantry **504** to rotate. As each of the rollers **512**A,B and **514** A,B is rotationally coupled with the gantry **504**, each of the rollers rotates with the gantry **504** regardless of whether or not it is a drive roller.

In the implementation shown in FIGS. **3***a-d*, there are four rotatable rollers configured and positioned to support the gantry **104**, however there may be more, or fewer. In the implementation shown, the four rollers comprise a plurality of 'front' drive wheels **512**A,B. These front wheels are at the T end of the gantry **504**. The plurality of drive wheels further comprises a plurality of 'rear' drive wheels **514**A,B. These rear wheels **514**A,B are at the G end of the gantry **504**. In other words, FIGS. **3***a-d* depict an implementation in which there are four drive wheels total, with two front drive wheels **512***a,b* and two rear drive wheels **514***a,b*. By positioning two rollers toward the front end of the gantry and two rollers toward the rear end of the gantry, the gantry is well-supported. As will be described elsewhere herein, at least one of the rollers is adjustable in order to mitigate or prevent a corkscrew effect, and having four rollers positioned as shown allows one or more of the rollers to be adjusted while ensuring the gantry **104** remains stable throughout the adjustment.

The radiotherapy apparatus **500** also comprises at least one support structure. The support structure(s) supports the rotatable rollers and holds them in place as they rotate. In FIGS. **3***a-d*, two support structures are shown: a front support structure **518**T toward the T end of the apparatus **500** which supports the front rollers **512**A and **512**B; and a rear support structure **518**G toward the G end of the apparatus **500** which supports the rear rollers **514**A and **514**B. The support structures **518**T,G may support the rollers via axles and axle holders (not shown in FIGS. **3***a-d*).

It will be appreciated that, by adjusting the position of one or more of the rollers **512**A,B and **514**A,B, it is possible to adjust the position of the gantry **504**. In particular, it is possible to find a position of the rollers **512**A,B and **514**A,B in which a corkscrewing movement of the gantry **504** is mitigated, or else eliminated entirely. One can think of this as adjusting the relative positioning of the gantry's rotational axis with respect to the respective rotational axes of each of the rollers **512**A,B and **514**A,B until there is no resultant force which can cause a corkscrewing effect.

FIG. **3***b* depicts a rear view of a configuration of the radiotherapy apparatus **504** in which the front rollers **512**A,B and the rear rollers **514**A,B are aligned with one another. In other words, the rotational axes of the roller **512**A and rear roller **514**A are substantially parallel and aligned with one another, and the the rotational axes of the roller **512**B and rear roller **514**B are substantially parallel and aligned with one another. In a mathematically ideal mechanical system, there should be no corkscrewing effect as the gantry **504** rotates in this configuration. However, due to mechanical stresses caused by heavy arm **534** and radiation source **530**, mechanical tolerances, motion and radial acceleration and deceleration, and other imperfections and deviations described elsewhere herein, a real apparatus in this configuration may still experience a corkscrewing effect as the gantry rotates on the rotatable support rollers **512**A,B and **514**A,B.

FIG. **3***c* depicts a rear view of a configuration of the radiotherapy apparatus **504** in which the rear wheels **514**A, B, have both been shifted in the B direction. FIG. **3***d* depicts a rear view of a configuration of the radiotherapy apparatus **504** in which the rear wheels **514**A,B have both been shifted in the A direction. By adjusting the position of the rear wheels in this way, it is possible to find a 'sweet spot' in which corkscrew motion is mitigated or else removed entirely. The sweet spot can be thought of as a position in which forces acting on the gantry along the G-T axis are eliminated, or reduced as much as possible.

Shifting the rear rollers **514**A, B in this manner, and/or by similarly shifting the front wheels **512**A,B, the position of the gantry rotation axis **516** in space may be adjusted. In particular, the gantry rotation axis **516** can be shifted with respect to the rotation axes of the rollers. By adjusting the relative position of the gantry rotation axis **516** with respect to the roller rotation axes, it is possible to find a "sweet spot" in which the cork-screwing effect is eliminated. In other words, by adjusting the relative positions of the rear wheels with respect to the front wheels, it is possible to find a configuration in which rotation of the gantry **504** does not result in the gantry **504** moving longitudinally along its rotation axis **516**. In this way, the radiation source **530** does not move along the T-G axis as the gantry **504** is rotated, and the radiation isocenter tolerance of the apparatus **500** is improved.

FIG. **4***a* and FIG. **4***b* depict a support structure **618** and a plurality of support rollers which are configured and positioned for supporting a gantry of a radiotherapy apparatus; for example, a radiotherapy apparatus as described above in relation to FIGS. **3***a-d*. FIG. **4***a* shows a rear view of the support structure **618**, and shows the support rollers positioned at the rear of the apparatus. The support structure **618** comprises two axle holders **630**A, **630**B which are displaced from one another along the A-B axis. In other words, the axle holders **630**A, **630**B are horizontally displaced from one another in a direction substantially perpendicular to the gantry rotation axis. The axle holders **630**A, **630**B each hold (i.e. support) an axle, around which a rotatable roller rotates. With respect to FIG. **4***a*, axle holder **630**A holds an axle about which roller **614**A rotates, and axle holder **630**B holds an axle about which roller **614**B rotates.

The apparatus depicted in FIG. **4***a* also comprises two idler wheels **615**A, **615**B. These wheels **615**A, **615**B form part of the plurality of rollers positioned underneath the gantry. The idler wheels are optional, and are positioned and configured to contact the gantry and provide additional support as the gantry rotates. The gantry drum itself is not shown in FIGS. **4***a*,**4***b* for increased clarity; however in use the drum of the gantry will contact each of the rotatable rollers.

FIG. **4***b* depicts the axle holder **630**A in greater detail. The axle holder **630**A comprises two axle holding elements **632**. The axle holding elements **631**A and **632**A are positioned either side of a rotatable roller **614**A. One axle holding element **632**A, T is located on the T side of the roller and the other axle holding element is located on the G side of the roller **632**A,G. The axle holding elements **631**A, **632**A each comprise an aperture through which the axle **650** may pass. The axle **650** may be referred to as a roller axle **650**. The roller **614**A and the axle **650** are configured such that the roller **614**A may be rotated around, and with respect to, the axle **650**. As such, the roller axle is a fixed axle, because, in use as the gantry rotates, the axle **650** does not rotate with the roller **614**A. The apparatus further comprises a wheel bearing **655** which enables the roller **614**A to rotate around the roller axle **650**.

While the axle **650** may be described as a fixed axle because it does not rotate with the roller **614**A as the gantry rotates, it may itself be rotated in order to adjust the position of the roller **614**A. The axle **650** itself has a rotation axis, and is configured such that rotation of the axle about the axle rotation axis displaces the rotatable roller in a direction substantially perpendicular to the gantry rotation axis. The form and function of the axle **650** will be described in greater detail with respect to FIGS. **5** and **6**.

FIG. **5** depicts a cross-section through the rotatable roller **614**A and roller bearing **660** depicted in FIGS. **4***a* and **4***b* in order to better depict roller axle **650**. Roller axle **650** is not depicted in cross-section, in order to provide a better visual aid to understanding its form and function.

The roller axle **650** is comprised of different sections along its length. The sections are substantially circular in cross-section, and have different radii. The roller axle **650** comprises a first section **651** about which the first roller **614**A rotates in use. The first section **651** may be referred to as an eccentric section, or an offset section. The first section is positioned eccentrically with respect to the first axle rotation axis. The first section **651** comprises a first radius. As can be appreciated from FIG. **5**, the first section **651** is the part of the axle **650** which is in contact with and coupled with the roller bearing **660**. The first section **651** and the roller bearing **660** share a common central axis, and this central axis defines the rotational axis of the roller **614**A. The first section **651** may be described as an eccentric or an offset section, as the first section is positioned eccentrically (offset) with respect to the rotation axis of the axle **650**. Accordingly, the rotational axis about which the roller **614**A is rotated is also offset with respect to the rotational axis of the axle **650**.

The roller axle **650** further comprises two sections having a second radius. These sections may be referred to as clamping sections. The clamping sections are optional. These clamping sections are positioned either side of the first section **651**, with a first clamping section **652**G being positioned toward the rear side of the apparatus compared with the other clamping section **652**T. The second radius of the clamping sections **652**T,G is larger than the first radius of the first section **651**, such that sections **652**T,G act to clamp the roller **614**A and roller bearing **660** in place around the first section **651** of roller axle **651**.

The roller axle **650** further comprises two sections having a third radius. These sections may be referred to as holding sections. These holding sections are positioned either side of the first section **651**, with one section **653**G being positioned toward the rear side of the apparatus compared with the other section **653**T. In implementations where the apparatus comprises clamping sections **652**G,T, the clamping sections **652**G,T are positioned between the first section **651** and the respective holding section **653**G,T. The third radius is less than the first and the second radii. Each holding section **653**T,G passes through an aperture in an axle holding element **632**. Each axle holding element **632** may be tightened or loosened, i.e. such that the aperture size can be increased or reduced. In ordinary use, the axle holding elements **632** are tightened to clamp the holding sections **653**T tightly in place such that the axle **650** is a fixed, stationary axle.

As described above, the first section **651** of the axle **650** is the part of the axle about which the roller **614**A is configured to rotate. The central axis of the first section **651** defines the rotational axis of the roller **614**A. This first section **651** of the axle **650** is offset with respect to the central axis of the holding sections **653** of the axle **650**. It is the holding sections **653** of the axle **650**, and their interaction with the axle holder, which define the rotational axis of the axle **650**. Accordingly, the rotational axis of the rotatable roller **614**A is offset with respect to the rotational axis of the axle **650**. Because the first section **651** is offset with respect to the axle rotation axis (as defined by the axle holder), the axle **650** is configured such that rotation of the axle **650** about the axle rotation axis displaces the rotatable roller **614**A. The axle **650** and its rotation axis is substantially parallel with the gantry rotation axis, and therefore rotation of the axle **650** causes the rotatable roller **614**A to be displaced in a direction substantially perpendicular to the gantry rotation axis.

Optionally, an axle cap **654** is provided at a distal end of the roller axle **650**. Such an axle cap can be used to facilitate rotation of the roller axle **650** when the axle holding elements **632** are loosened. The axle cap may comprise, for example, a nut, a bolt, or both, and may be substantially hexagonal in shape (as shown in FIG. **5**).

When the radiotherapy apparatus is installed on site, for example at a hospital, or else when the apparatus is undergoing maintenance, the axle holding elements **632** may be loosened and the axle **650** may be rotated about its rotational axis in order to adjust the position of the rotatable roller **614**A. The axle **650** can be rotated via a rotation of the axle cap **654**, for example via a wrench, spanner or a specialised tool configured to interact with the axle holder.

FIG. **6** depicts the axle **650** as described above in relation to FIGS. **4***a*, **4***b*, and **5**. As noted above, the roller axle **650** is comprised of different sections along its length which have different radii. The axle **650** takes the form of an eccentric shaft, comprising an eccentric section position substantially centrally along the length of the shaft, and about which a roller will rotate in use.

The roller axle **650** includes the eccentric section **651** around which the roller will rotate in use, the clamping sections **652**T,G, the holding sections **653**T,G, and an axle cap **654**. When positioned in the axle holder, the axle **650** is rotatable about an axle rotation axis defined by the central axis of the holding sections **653**T,G. The holding sections **653**T,G are configured to pass through the axle holder, and in particular to pass through the axle holding elements **632** of the axle holder, thereby defining the rotational axis for the axle **650**.

As can be appreciated from the figures, the first section **651** is eccentric, i.e. offset, with respect to the holding sections **653**T,G. Equivalently, the first section **651** is eccentric, i.e. offset, with respect to the axle rotation axis. As such, the rotational axis of the roller will also be offset from the axle rotational axis.

The first section **651** is not positioned centrally with respect to the axle rotational axis. In other words, the central axis of the first section **651** does not align with the axle **650** rotational axis. The first section **651** is substantially cylindrical, and its central axis may be parallel with the axle **650** axis of rotation. In use, the axle **650**, each of its sections and the rotation axis of the axle **650** are positioned substantially parallel with the gantry rotation axis.

The clamping sections **652**T,G are shown here, but are optional. These clamping sections **652**T,G share a common central axis and are also offset with respect to the rotation axis of the axle **650**, to the same degree and in the same manner as the first section **651**.

The axle cap **654** is located at the G end of the axle **650**. The axle cap **654** provides the means by which the axle **650** can be rotated around its rotational axis. The axle cap **654** may differ in radius from at least the second section **653**G of the axle **650**. The central axis of the axle cap **650** aligns with the axle **650** rotation axis. As such, the rotation of the axis **650** via the axle cap **654** rotates the axle **650** about its rotational axis.

The offset nature of the first section **651** of the axle **650** causes the first section **651** to be rotated about a point other than its geometrical centre. As the first rotatable roller rotates about first section **651** when in use, a rotation of the axle **650** causes a displacement of the rotatable roller in any radial direction relative to the axle **650** rotation axis. Thus, the axle **650** is configured such that, by rotating the axle **650** about the axle **650** rotation axis, the rotatable roller may be displaced in any radial direction relative to the axle rotation axis.

FIGS. **7***a*-*d* are schematics depicting a rear view of a roller **714** configured to rotate about an axle which comprises a first section **753** and a holding section **753**. The optional clamping sections of the axle are not shown here for clarity. FIGS. **7***a*-*d* depict the same arrangement, with the axle having been rotated successively through 90°. In each of FIGS. **7***a*-*d*, the axle's centre of rotation **760** and the roller's centre of rotation **770** are depicted. As an example, the axle may be configured such that the distance between these axes of rotation is approximately 5 mm.

In FIG. **7***a*, the roller **714** is at its maximal displacement in the u direction. Arrow U_1 depicts the distance from the axle axis of rotation **760** to the outer circumference of the roller **714** in the U direction. FIG. **7***b* depicts the same roller **714** and axle, where the axle has been rotated about its axis of rotation **760**. The roller **714** is now at its maximal displacement n the b direction. Arrow U_2 depicts the distance from the axle axis of rotation **760** to the outer circumference of the roller **714** in the u direction. It will be appreciated from inspection of FIG. **7***b* that rotation of the axle 90° about its rotation axis **760** has caused the roller **714** to move in the d and b directions, i.e. in a direction perpendicular with the radiotherapy axis of rotation. Accordingly, $U_1>U_2$. The roller axis of rotation **770**, located along the central axis of the first, central section of the axis **751**, has also been displaced with respect to the axle axis of rotation, and in particular has been rotated in a clockwise direction with respect to the configuration depicted in FIG. **7***a*.

FIGS. **7***c* and **7***d* depict the roller **714** in its maximal displacement in the d and a directions respectively. $U_2>U_3$. Because the first section **760** of the axis (around which the roller **714** rotates) is offset with respect to the section of the axis which defines the axle rotation axis **760**, it can be appreciated that rotating the axle has the effect of rotating the roller axis of rotation **770** about the axle's axis of rotation **760**. In other words, rotating the axle causes the roller to be displaced in a direction perpendicular to the axle rotation axis. Because the axle is positioned substantially parallel with the gantry rotation axis, the displacement direction is also substantially perpendicular to the gantry rotation axis. By rotating the axle about its rotation axis **760** in a first rotational direction (for example clockwise, as is depicted moving in order through FIGS. **7***a*-*d*), the first roller rotation axis **770** is also rotated in the first rotational direction.

The apparatus disclosed herein thus provides a means of mitigating the 'corkscrew effect' via repositioning or adjustment of one or more rotatable rollers which support the apparatus. By implementing the present disclosure, any of the rollers supporting the gantry can be made independently adjustable in space to provide adjustment of the gantry. By adjusting one or more of the rollers in the a,b,u,d directions, a sweet spot can be found in which the corkscrew effect is reduced or eliminated entirely. The adjustment is mechanically simple, meaning the adjustment means is unlikely to fail or require repair. The use of eccentrics in the manner described minimises the complexity of adjustment, maintains the axle axes on a horizontal plane, and provides the ability to correct any minor error in the gantry rotational axis (i.e. to bring it back into alignment with the horizontal).

FIG. **8** depicts a radiotherapy apparatus **700** according to the present disclosure having a plurality of support wheels, including two stationary front wheels **712**A and **712**B and two adjustable rear wheels comprising an A-side rear wheel **714**A and a B-side rear wheel **714**B (not shown in the figure). A mechanical arm **534** extends from the gantry **754** toward housing **730**. While not shown here, the housing **730** is configured to hold the radiation source and/or the beam generation system.

The front wheels **712**A,B are configured to rotate about their respective axles, which are each held in place by respective axle holders on the under-gantry support structure **718**. The front wheels **712**A,B are displaced from one another along the A-B axis, and the rear wheels **712**A,B are also displaced from one another along the A-B axis. The front wheels **712**A, B are displaced from the rear wheels **714**A,B along the T-G axis. Any single wheel or combination of the front or rear wheels may be drive wheels, however in this implementation each of the front and rear wheels is a drive wheel (drive means such as rotary motors not shown in the figure). The gantry further comprises two rims around its circumference, where the front wheels **712**A,B support the gantry **754** via contact with the front rim **780**T, and where the rear wheels **714**A,B support the gantry **754** via contact with the rear rim **780**G. By driving each of the wheels together, the angle of rotation of the gantry **754** (and hence the radiation source) can be controlled.

The rear wheels **714**A,B are both adjustable via the means described above. In particular, the apparatus **700** comprises rear axles about which the rear wheels spin, and each rear axle is configured such that rotation of the axle about its rotation axis displaces its wheel in a direction substantially perpendicular to the T-G axis.

Accordingly, it will be appreciated that the apparatus **700** comprises a plurality of rollers **712**A, **712**B, **714**A, and **714** B (not shown in the figure). The apparatus comprises a first axle about which a first roller **714**A rotates, and the first axle itself is configured to rotate about a first axle rotation axis. The first axle is configured such that rotation of the first axle about the first axle rotation axis displaces the first rotatable roller **714**A in a direction substantially perpendicular to the gantry rotation axis, in the manner described above with respect to FIGS. **4***a* to **7***d*. A second rotatable roller **714**B of the plurality of rotatable rollers is configured to rotate about a second roller axle. The second axle is configured such that rotation of the second axle about the second axle rotation axis displaces the second rotatable roller **714**B in a direction substantially perpendicular to the gantry rotation axis, in the manner described above with respect to FIGS. **4***a* to **7***d*. The first and second axles are horizontally displaced, i.e. separated, with respect to each other along the A-B axis, i.e. along an axis substantially perpendicular to the gantry rotation axis. This displacement is achieved by virtue of the support structure **718** comprising a first axle holder configured to hold the first axle and a second axle holder configured to hold the second axle, where the first and second axle holders are displaced from one another along the A-B axis.

Upon installation of the apparatus **700**, the drive wheels are driven in order to rotate the gantry **754**, and the gantry is monitored for a corkscrewing movement in a direction along the T-G axis. If the gantry **754** is exhibiting signs of the corkscrewing effect, then one or both of the rear wheels may be adjusted. Rotation of the gantry can then be checked again for evidence of the corkscrewing effect. The process can be repeated until the effect has been sufficiently mitigated, or eliminated entirely.

Accordingly, disclosed herein is a method, which may be performed for example upon installation or during routine repair and servicing of a radiotherapy device. The method involves displacing a first roller of a plurality of rotatable rollers positioned underneath a gantry of a radiotherapy device. The radiotherapy device may be a radiotherapy device in accordance with any of the implementations disclosed herein, and comprises: rotating a first axle about its rotation axis to displace a first rotatable roller in a direction substantially perpendicular to the gantry rotation axis. The method may further comprise effecting a rotation of the gantry, monitoring for linear movement of the gantry in a direction substantially parallel with its rotation axis, and adjusting/displacing the first rotatable roller based on any detected motion during the monitoring. This may be an iterative process comprising several rounds of monitoring and adjustment until the corkscrew effect has been sufficiently mitigated.

Because both the rear wheels **714**A,B may be displaced/moved/adjusted, the entire rear face of the gantry can be adjusted. For example, by displacing both rear wheels **714**A,B in the A direction, the entire rear rim **718**G and rear face of the gantry is moved in the A direction. This type of adjustment is the type of adjustment depicted in FIGS. **3***a-d* and described above. The advantage of this arrangement, i.e. translating the rear of the gantry in this way using two independently adjustable rollers, means that the corkscrew effect can be mitigated while the height of the gantry rotational axis can be kept reasonably constant throughout the adjustment, the gantry rotational axis is kept horizontal throughout the adjustment, and maintains the perpendicular relationship between the radiation beam and the gantry rotational axis. By providing an eccentric cam configuration on both the rear wheels, or equivalently on both the front wheels, they can be arranged in such a way that when both eccentrics are rotated to translate the gantry substantially in a lateral direction, one eccentric lifts the gantry slightly while the other eccentric lowers the gantry slightly. The resultant lateral shift in the height is negligible, meaning the horizontal axis remains substantially unmoved and horizontal. The corkscrewing effect can therefore be mitigated or eliminated, without changing the position of the isocenter and hence negatively impacting the accuracy of radiotherapy.

Adjusting both the rear wheels may result in a very small displacement of the x-ray source laterally. According to implementations of the present disclosure, this small displacement can be accommodated in two ways, for example by laterally adjusting or positioning the patient support system or a gantry support system to counter the displacement; or by adding a further adjustment to the other set of wheels (in this case front wheels) to move them in substantially the opposite direction. Thus, the adjustment at front and rear could be designed to give a zero lateral displacement but still displace the rear wheels laterally with respect to the front wheels and have the desired effect on minimizing or reducing corkscrewing.

It is to be understood that the above description is intended to be illustrative, and not restrictive. Many other implementations will be apparent to those of skill in the art upon reading and understanding the above description. Although the present disclosure has been described with reference to specific example implementations, it will be recognized that the disclosure is not limited to the implementations described, but can be practiced with modification and alteration within the spirit and scope of the appended claims. Accordingly, the specification and drawings are to be regarded in an illustrative sense rather than a restrictive sense. The scope of the disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

In this document, the terms "a" or "an" are used, as is common in patent documents, to include one or more than one, independent of any other instances or usages of "at least one" or "one or more." In this document, the term "or" is used to refer to a nonexclusive or, such that "A or B" includes "A but not B," "B but not A," and "A and B," unless otherwise indicated. In the appended claims, the terms "including" and "in which" are used as the plain-English equivalents of the respective terms "comprising" and "wherein." Also, in the following claims, the terms "including" and "comprising" are open-ended, that is, a system, device, article, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms "first," "second," and "third," etc. are used merely as labels, and are not intended to impose numerical requirements on their objects.

What is claimed is:

1. A radiotherapy apparatus comprising:

a gantry rotatable about a gantry rotation axis; and

a plurality of rotatable rollers positioned underneath the gantry and configured to support the gantry, wherein a first rotatable roller of the plurality of rotatable rollers is configured to rotate about a first axle, wherein the first axle is configured to rotate about a first axle rotation axis, and wherein the first axle is configured such that rotation of the first axle about the first axle rotation axis displaces the first rotatable roller in a direction substantially perpendicular to the gantry rotation axis.

2. The radiotherapy apparatus of claim **1**, further comprising:

a support structure comprising a first axle holder configured to support the first axle and define the first axle rotation axis.

3. The radiotherapy apparatus of claim **2**, wherein the first axle comprises one or more different sections along its length, including an eccentric section positioned eccentrically with respect to the first axle rotation axis, the eccentric section having a different radius compared to other sections of the axle, and wherein the first rotatable roller is configured to rotate about the eccentric section.

4. The radiotherapy apparatus of claim **3**, wherein the eccentric section is offset with respect to the first axle rotation axis.

5. The radiotherapy apparatus of claim **3**, wherein the eccentric section has a central axis, wherein the central axis is parallel with, but does not align with, the first axle rotation axis.

6. The radiotherapy apparatus of claim **3**, wherein the eccentric section defines a first roller rotation axis about which the first rotatable roller rotates, wherein the first roller rotation axis is displaced from the first axle rotation axis.

7. The radiotherapy apparatus of claim **6**, wherein the first axle is configured such that, by rotating the first axle about the first axle rotation axis in a first rotational direction, the firs roller rotation axis is rotated in the first rotational direction.

8. The radiotherapy apparatus of claim 3, wherein the first axle further comprises:

two or more holding sections positioned on either side of the eccentric section, wherein the two or more holding sections pass through the first axle holder.

9. The radiotherapy apparatus of claim 8, wherein the two or more holding section are substantially cylindrical, and wherein the two or more holding sections have a shared central axis aligned with the first axle rotation axis.

10. The radiotherapy apparatus of claim 1, wherein the plurality of rotatable rollers comprises at least one drive roller configured to drive rotation of the gantry.

11. The radiotherapy apparatus of claim 1, wherein the plurality of rotatable rollers further comprises a second rotatable roller, wherein the second rotatable roller is configured to rotate about a second roller axle, wherein the second roller axle is configured to rotate about a second axle rotation axis, and wherein the second roller axle is configured such that rotation of the second roller axle about the second axle rotation axis displaces the second rotatable roller in a direction substantially perpendicular to the gantry rotation axis.

12. The radiotherapy apparatus of claim 11, wherein the first axle and the second roller axle are horizontally displaced with respect to each other along an axis substantially perpendicular to the gantry rotation axis.

13. The radiotherapy apparatus of claim 11, wherein the first rotatable roller and the second rotatable roller are both positioned either toward a rear of the gantry or toward a front of the gantry.

14. A method for displacing a first rotatable roller of a plurality of rotatable rollers positioned underneath a gantry of a radiotherapy device, wherein each roller in the plurality of rotatable rollers is configured to support the gantry, wherein the gantry is rotatable about a gantry rotation axis, wherein the first rotatable roller of the plurality of rotatable rollers is configured to rotate about a first axle, wherein the first axle is configured to rotate about a first axle rotation axis, and wherein the first axle is configured such that rotation of the first axle about the first axle rotation axis displaces the first rotatable roller in a direction substantially perpendicular to the gantry rotation axis, the method comprising:

rotating the first axle about the first axle rotation axis to displace the first rotatable roller in the direction substantially perpendicular to the gantry rotation axis.

15. The method of claim 14, wherein the first axle includes an eccentric section, wherein the eccentric section is positioned eccentrically with respect to the first axle rotation axis, and wherein the first rotatable roller rotates around the eccentric section.

16. A radiotherapy device comprising:

a gantry;

one or more rotatable rollers positioned underneath the gantry, wherein the gantry is rotatable about a gantry rotation axis, wherein the one or more rotatable rollers are configurable to support the gantry, wherein a first rotatable roller of the one or more rotatable rollers is configurable to rotate about a first axle, wherein the first axle is configurable to rotate about a first axle rotation axis, wherein the first axle rotation axis is configured such that rotation of the first axle about the first axle rotation axis displaces the first rotatable roller in a direction substantially parallel to the gantry rotation axis, and wherein a second rotatable roller is configurable to rotate about a second roller axle, wherein the second roller axle is configured to rotate about a second axle rotation axis, and wherein the second roller axle is configured such that rotation of the second roller axle about the second axle rotation axis displaces the second rotatable roller in a direction substantially perpendicular to the gantry rotation axis; and

a support structure, the support structure including a first axle holder configurable to support the first axle and define the first axle rotation axis.

17. The radiotherapy device of claim 16, wherein the first axle includes an eccentric section, wherein the eccentric section is positioned eccentrically with respect to the first axle of rotation, wherein the eccentric section is offset with respect to the first axle rotation axis, and wherein the first rotatable roller is configurable to rotate about the eccentric section.

18. The radiotherapy device of claim 17, wherein the eccentric section defines a first roller rotation axis about which the first rotatable roller rotates, wherein the first roller rotation axis is displaced from the first axle rotation axis, and wherein the first axle is configured such that, by rotating the first axle about the first axle rotation axis in a first rotational direction, the first roller rotation axis is rotated in the first rotational direction.

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